Optical matching and beam loss in a booster-to-storage-ring transfer line for nonlinear-kicker injection at the Korea Light Source

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Abstract

A booster-to-storage-ring (BTS) transfer line must deliver the extracted booster beam within the optical and geometric constraints of storage-ring injection. We study a design version of the Korea Light Source BTS line with three bending magnets and nine quadrupoles, using the injection-septum geometry of a nonlinear-kicker injection study. Once the line is matched to the prescribed exit lattice functions, the nominal beam size at both ends is fixed, so the nominal clearance of the injection-septum blade, the active restriction, is the same for every matched setting, and lowering the interior peak beta changes only interior clearances. The quadrupoles act on the blade loss through the trajectory. After static correction with field trims of existing dipoles, booster shot jitter dominates, and for matched ends its partition between exit position and angle depends only on the horizontal phase advance. A Gaussian loss model validated against particle tracking shows that the loss of a matched setting depends on the phase advance alone and is smallest at the lower edge of the matched range. There, the best-found setting reduces the tracked loss by a factor of 3 to more than 10 on independent error realizations, at the cost of larger exit-angle jitter. This holds at a stress level of booster jitter, 1.3 and 2.2 beam sizes in position and angle; at 0.1–0.2 of that level the blade loss is negligible for every setting, and the design output of the line is the position–angle partition delivered to the injection kicker. A dispersion-free exit extends that partition. Storage-ring capture is outside the scope of the study.

Keywords: beam dynamics; error analysis; Korea Light Source; transfer line

1 Introduction

Diffraction-limited storage rings require control of the beam distribution delivered to the injection region [1]. The booster-to-storage-ring (BTS) transfer line connects the extracted booster beam to the prescribed injection conditions of the storage ring. Its optical design must accommodate the incoming emittance and momentum spread, satisfy the exit lattice functions, and maintain adequate clearance through the magnets and injection septum. At the septum, the beam envelope and trajectory must fit within a channel bounded by the blade; downstream, both position and angle affect the conditions delivered to the injection kicker. The 4 GeV Korea Light Source (KLS), described in its technical design report (TDR) [2] and in Refs. [3, 4], provides the application considered here. In the TDR design, the BTS line consists of two dipole and eight quadrupole magnets, and the storage ring is filled by off-axis injection with a four-kicker orbit bump and an in-vacuum thin septum with a 1 mm blade [2]. The line studied here is a different design version of the BTS, with three main dipoles and nine quadrupoles (Section 2), and its injection geometry

is taken from our study of nonlinear-kicker magnet (NKM) injection into the KLS storage ring [5]; the results apply to this version of the line.

Previous transfer line studies establish the importance of treating the accelerator interfaces and available geometry together. The HEPS high-energy transport-line design matches the injection-point lattice functions while accommodating the relative positions of the booster and storage ring, equipment clearances, and the selected injection scheme [6]. A subsequent HEPS study updates the transfer lines in response to the evolving accelerator layouts and optics [7]. For Diamond-II, the transfer lines are designed to preserve bunch properties with low loss while retaining diagnostic capability and flexibility for matching; the BTS modifications connect the new booster extraction geometry to the storage-ring septum [8]. The ESRF–EBS rematching similarly includes injection-region fields and mechanical constraints [9]. These examples show why an optical target and the physical restrictions along the transport path must be specified together.

Beam characterization and trajectory control address different parts of this matching problem. The SLS 2.0 transfer line includes a nondispersive section for quadrupole-scan measurements and a corrector configuration for separate control of injection position and angle [10]. Its commissioning results connect beam characterization to downstream matching and trajectory stabilization. The Sirius injection study reports efficiencies separately for the injector stages and examines booster realignment, emittance exchange, and remaining reproducibility issues [11]. Separating these stages prevents a transfer-line improvement from being equated with the same change in storage-ring injection efficiency. It also motivates evaluating a nominally matched line under incoming beam fluctuations and residual trajectory errors, rather than relying on envelope plots alone.

The requirements imposed by the injection device provide an additional design constraint. Pulsed-sextupole injection at MAX IV relates the incoming beam distribution to the nonlinear injection process [12]. A recent TPS study proposes a new booster-to-storage-ring line for nonlinear-kicker injection and identifies existing-ring geometry, trajectory correction, and injected-beam characterization as design considerations [13]. Together, these studies motivate distinguishing nominal optical matching, clearance at the limiting restriction, and sensitivity to the position and angle delivered to the injection system. The present study examines those distinctions within a fixed BTS model with explicit aperture and jitter models.

A common transfer-line design objective is to reduce the peak beta functions and thereby increase aperture margin. For fixed emittances, momentum spread, and matched entrance and exit lattice functions, however, quadrupole rematching preserves the nominal beam sizes at the boundaries. Restrictions in the fixed sections upstream and downstream of the quadrupoles can therefore limit the line even when its interior envelopes are reduced. The same boundary conditions leave freedom in the betatron phase advance, which changes the transport of booster position and angle jitter. A matched setting can consequently change the loss probability at a septum blade without changing its nominal clearance. Distinguishing these mechanisms is necessary to determine whether a proposed optics improvement addresses the limiting restriction or only redistributes margin elsewhere in the line.

The goal of this study is to identify which properties of a matched BTS setting control the beam loss at the active restriction, and what a designer can choose among matched settings. The baseline setting is the nine-quadrupole configuration supplied with the line before matching; it does not satisfy the prescribed exit and serves only as the starting point. Bounded optical matching followed by vertical-beta optimization gives the optimized low-beta setting (low beta). A matched intermediate setting, referred to below as the matched reference, is the comparison for all error studies. The line is modeled with element-wise apertures and the injection-septum blade geometry of the NKM injection study [5]. The subsequent error study compares the matched reference, low-beta, and loss-based settings after static trajectory correction using field trims of existing dipoles. A Gaussian loss probability derived from the transported beam and centroid

covariance is validated against particle tracking and used to select a loss-based setting before evaluation on independently seeded error realizations. The analysis also examines the phase advance of the matched settings, a dispersion-free exit target, and the dependence on the error levels, in particular on the booster jitter. The study ends at the transfer-line exit; storage-ring capture and NKM transparency require the subsequent injection calculation.

Section 2 defines the model, Section 3 the criteria and their validation, Section 4 the results, and Section 5 the design implications and limitations.

2 Parametric model and physical observables

2.1 Lattice, coordinates, and boundary conditions

The baseline BTS model contains 36 elements over 21.789 m, from marker m1 at the entrance to marker m4 downstream of the injection septum. Beam dynamics simulations use the Accelerator Toolbox (AT) [14] through its Python implementation pyAT [15]. The line has nine 0.2 m quadrupoles in three triplets, three main dipole bending magnets (`b1`, `b2`, and `b3`, in beam order), the booster extraction kicker `kext`, the extraction septum `sept_in` and the injection septum `sept_ex`; the kicker and both septa are modeled as dipole elements (Fig. 1). Quadrupoles are labeled `qij`, where i identifies the triplet and j the magnet within that triplet, both in beam order. Thus `q11` is the first quadrupole in the first triplet and `q33` is the last quadrupole in the third triplet. Local coordinates are $(x, p_x, y, p_y, \delta, ct)$ with positions/path length in m, normalized canonical momenta approximately angles in rad, and $\delta = \Delta p/p_0$ dimensionless. The six design bend angles are 7.5 mrad, 88.5 mrad, −111.701 mrad, 176 mrad, −111.701 mrad and 88.5 mrad; reference geometry is held fixed when fields or particle momenta change.

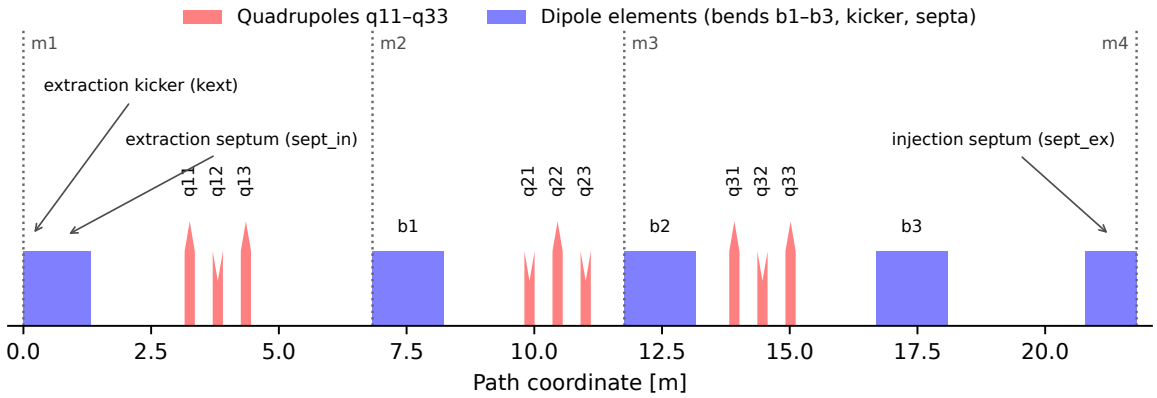


Figure 1: Baseline BTS element sequence drawn with Accelerator Toolbox synoptic plotting. Red symbols are the quadrupoles `q11`–`q33`; blue blocks are the dipole elements: the extraction kicker `kext`, the extraction septum `sept_in`, the main bends `b1`–`b3` and the injection septum `sept_ex`. Blank intervals are drifts. Dotted lines mark the markers m1–m4 used for boundary and intermediate optics observations.

Entrance and exit lattice functions (Table 1) are inherited design values. The exit m4 is the injection-septum exit: the NKM injection study uses exactly these values as its injected beam and transports it 2.26 m to the kicker. The configured nonzero exit dispersion (80.9 mm, 47.5 mrad) differs from the ring dispersion at the injection kicker, which is about 0.3 mm; we therefore also examine a dispersion-free exit. Quadrupole family bounds, bore radius, pole-field limit, the 60 m beta screening ceiling and square aperture limits are model parameters. The BTS line is independently validated using analytical maps of the same AT element sequence.

Table 1: Prescribed entrance and exit lattice functions. The exit values describe the injected beam at the injection-septum exit used in the NKM injection study [5]. A dispersion-free exit is examined in Section 4.4.

Observable	m1 entrance	m4 target
β_x, β_y [m]	7.560000, 12.269000	2.336495, 4.256241
α_x, α_y [1]	1.523100, −1.654700	−0.016335, 0.017772
D_x [m], D'_x [1]	0.276200, −0.065700	0.080868, 0.047472

The four-station comparison model places square apertures at markers m1–m4 (19.35 mm at the line ends, 30 mm at two interior markers) and none on the magnets. That four-station model provides a comparison with the element-wise restrictions of Section 2.3.

2.2 Beam covariance and clearance

This study uses the booster horizontal design input $\epsilon_x = 7.67\,\mathrm{nm\,rad}$ and $\sigma_\delta = 0.106\,\%$ [4]. The vertical input $\epsilon_y = 0.767\,\mathrm{nm\,rad}$ corresponds to a vertical-to-horizontal emittance ratio of 10 %. Betatron coordinates and momentum are independent untruncated Gaussians before dispersion is added; initial transverse coupling is zero and $ct = 0$ for this RF-free single pass. The transverse blocks of the complete covariance are

$$\Sigma_{u,\mathrm{bet}} = \epsilon_u \begin{pmatrix} \beta_u & -\alpha_u \\ -\alpha_u & (1+\alpha_u^2)/\beta_u \end{pmatrix}, \qquad \Sigma_0 = \Sigma_{\mathrm{bet}} + \sigma_\delta^2 \mathbf{d}\mathbf{d}^{\mathsf{T}}, \tag{1}$$

where $u \in \{x, y\}$ denotes the transverse plane, β_u and α_u are the Courant–Snyder parameters and ϵ_u the emittance in that plane, Σ_{bet} is the 6×6 betatron covariance with the two 2×2 blocks $\Sigma_{u,\mathrm{bet}}$ on its diagonal, and $\mathbf{d} = (D_x, D'_x, D_y, D'_y, 1, 0)^{\mathsf{T}}$. The same index u is used for the plane in all equations below. Rolls of the quadrupoles about the beam axis (Section 2.4) couple the horizontal and vertical planes downstream of each rolled magnet. Full covariance transport, $\Sigma_s = M_s \Sigma_0 M_s^{\mathsf{T}}$, retains that coupling and momentum correlation. A local affine map gives the centroid $\bar{\mathbf{z}}_s = M_s \bar{\mathbf{z}}_0 + \mathbf{b}_s$.

For each aperture station s and plane u, with lower and upper aperture edges a_- and a_+, we define

$$c_u(s) = \min[a_+ - \bar{u}(s), \bar{u}(s) - a_-] - 5\sqrt{\Sigma_{uu}(s)}, \qquad c_{\min} = \min_{s \in \mathrm{stations}, u \in \{x,y\}} c_u(s). \tag{2}$$

Here $\bar{u}(s)$ is the beam centroid and $\sqrt{\Sigma_{uu}(s)}$ the projected rms beam size, including the dispersive contribution. The first term in Eq. (2) is the distance from the centroid to the nearer aperture edge; subtracting five rms beam sizes gives the margin $c_u(s)$ that remains between the five-sigma beam envelope and that edge. A positive c_u means that the envelope fits inside the aperture, $c_u = 0$ that it touches the edge, and a negative c_u that it overlaps the edge by $|c_u|$. The minimum $c_{\min}$ over all stations and both planes identifies the limiting restriction. The five-sigma envelope is a conventional coverage measure of the projected distribution; an untruncated Gaussian beam is never completely contained. Because the beam size scales as $\sqrt{\beta_u}$, a reduction of the beta function by some metres does not increase the clearance by a comparable length: the clearance changes only by five times the change of the rms size at the limiting station, and not at all where the optics are fixed by the boundary conditions (Section 3).

The quadrupole strengths are bounded so that every setting considered remains within the field range of conventional quadrupole magnets. At 4 GeV the magnetic rigidity is $B\rho = p_0/e \approx 13.342\,56\,\mathrm{T\,m}$. A normalized strength K in m^{-2} corresponds to the field gradient $G = K(B\rho)$ in $\mathrm{T\,m}^{-1}$, and a quadrupole with bore radius r then has the pole-tip field $B_{\mathrm{pole}} = |G|r$ in T. The strength bounds are $|K| \le 2.5\,\mathrm{m}^{-2}$ for the outer quadrupoles q11 and q33 and $|K| \le 3\,\mathrm{m}^{-2}$ for

the other seven. With $r = 19.35\,\mathrm{mm}$, the largest gradients are $33.356\,\mathrm{T\,m^{-1}}$ and $40.028\,\mathrm{T\,m^{-1}}$, and the corresponding pole-tip fields are 0.645 T and 0.775 T, respectively. Both remain below the 1.2 T pole-tip limit used for screening. This check confirms that the optimized strengths are consistent with the magnet model; it is not an engineering design of the magnets.

2.3 Element-wise restrictions

Every element of the BTS line carries its own aperture (Table 2), and AT checks each aperture at the element entrance; the line end (m4) carries the injection-septum limits, so the blade is checked at both ends of the septum. Particle losses are therefore counted at these declared check points, not as continuous interception along a chamber or channel. Interior envelopes are sampled for the nominal profile on an observation-only copy of the line, with four subdivisions per quadrupole and eight per bend or septum; these nominal samples locate the interior minimum clearance but do not certify the beam tails under errors.

The injection-septum blade separates the main-field channel traversed by the injected beam from the stored-beam region, where a small magnetic field is required. Its geometry follows the NKM injection study [5]; for comparison, the baseline KLS injection uses an in-vacuum thin septum with a 1 mm blade [2]. Let X denote the horizontal coordinate relative to the storage-ring reference orbit at the septum exit. The blade face toward the stored beam is at $X = -16\,\mathrm{mm}$ and its thickness is 2 mm, so the face toward the injected beam is at $X_b = -18\,\mathrm{mm}$. The nominal injected-beam centroid is at $X_{\mathrm{inj}} = -19.478\,\mathrm{mm}$, giving the centroid-to-blade distance

$$d_b = X_b - X_{\mathrm{inj}} = 1.478\,\mathrm{mm}. \tag{3}$$

For the configured beam, this distance is approximately three rms horizontal beam sizes plus 1 mm. In the local BTS coordinate at the exit, $x = X - X_{\mathrm{inj}}$, the nominal injected beam is centered at $x = 0$ and the blade face is at $x = +d_b$. This boundary is applied on the side facing the stored beam in the injection septum and at the line end (marker m4). The five-sigma clearance in Eq. 2 additionally subtracts the projected beam envelope and includes any centroid displacement; it is distinct from d_b.

The injected beam is intentionally deflected by the septum's main magnetic field while it travels through the injection channel. Proximity to the blade does not imply that the beam is in a field-free region. In the BTS model, the septum's bending field is represented by a dipole element, while the blade face is represented separately by the aperture boundary. Detailed spatial fringe-field and leakage-field distributions are not resolved. The blade dimensions and offset are study parameters rather than established hardware specifications; the other aperture dimensions are model parameters, and the bend gap and the extraction-septum aperture are swept (Section 4).

Bend chambers are taken as curved: the sagitta of the reference arc over its chord is 30.8 mm in b2 and 19.5 mm in b1/b3, so a straight chamber would need that much extra width. Clearances (Eq. 2) are evaluated at every aperture, with the projected halfwidth of a round aperture reduced by the other plane's centroid.

2.4 Errors, jitter and trajectory correction

Static machine errors are independent Gaussian errors of the nine quadrupoles, namely gradient errors (0.1 % rms), x/y offsets (100 µm rms) and rolls about the beam axis (0.5 mrad rms), together with field errors of the six dipole elements (0.05 % rms). Each independent draw of these static errors defines one error realization of the line. Each shot adds booster jitter at m1: entrance beta factors (5 %), centroid position 0.5 mm, angle 0.2 mrad and momentum 0.1 %, the latter carried by the entrance dispersion. These levels are model inputs; the booster jitter is the same as in the NKM injection study [5]. In beam-size units the betatron jitter is large: at m1 the rms beam is 0.379 mm and 91 µrad, so the position and angle jitter are 1.3 and 2.2 beam sizes,

Table 2: Element-wise aperture model, reported as halfwidths. For the injection septum, a_x is the nominal injected-centroid-to-blade distance on the $+x$ side, not the blade thickness or its offset from the storage-ring reference orbit. The outer wall is at -8.5 mm in local BTS coordinates.

Region	Shape	a_x [mm]	a_y [mm]
Drifts	round	19.350	19.35
Quadrupoles q11–q33	round	19.350	19.35
Extraction kicker	rect.	15.000	10.00
Extraction septum	rect.	7.500	5.00
Bends b1–b3 (curved)	rect.	30.000	10.00
Injection septum	one-sided	1.478	5.00

and the mean centroid action is 54 times the emittance. Booster extraction stability is commonly specified as a fraction of the beam size; no KLS specification is available to us, so we treat these values as a stress level and report the dependence of the results on it. The sensitivity to all error levels is examined in Section 4.5.

Static trajectory errors are corrected without changing the line: no corrector or monitor element is added. The horizontal knobs are field trims of the six existing dipoles, applied as additional fixed-geometry dipole fields like the dipole errors. Horizontal readings are taken at existing element boundaries, after each quadrupole, at the injection-septum entrance and at the line end, with a static 50 µm rms reading error (offset and calibration residual). The lattice model contains no monitor elements, so these eleven readings are assumed instrumentation, and the six trims are assumed to be independently powered; fewer readings and series-powered bends are examined in Section 4.2. The trims minimize the weighted readings (septum entrance and end weighted 3) with the response of the error-free model line. With $\mathbf{r}$ the vector of readings, A the response matrix of the readings to the trims $\boldsymbol{\theta}$, and W the diagonal weight matrix, the trims minimize

$$\|W(\mathbf{r} + A\boldsymbol{\theta})\|^2 + \lambda^2\|\boldsymbol{\theta}\|^2. \tag{4}$$

The second term is a Tikhonov (ridge) regularization [16]: it penalizes large trims and keeps the solution stable when the response matrix is ill-conditioned. Here λ is 10^{-3} times the largest singular value of WA. The trims are limited to ± 1 mrad of kick, and the correction is applied in two passes. The correction uses no knowledge of the actual errors and does not act on shot jitter. The line has no vertical knob, so the vertical plane is not corrected. Every error realization is evaluated uncorrected and corrected on identical particles. When settings are compared, the static magnet errors, shot jitter and particles are shared, whereas the reading errors are drawn separately for each setting; paired comparisons therefore include the effect of different reading errors.

3 Physical criterion, validation and comparison protocol

3.1 Active restrictions and boundary-fixed clearances

The two end stations sit where the optics are fixed by boundary data: the entrance Courant–Snyder parameters and dispersion are inputs, and every admissible setting matches the exit to its target. For a centered nominal beam the clearance there is therefore $c_u = a - 5\sqrt{\epsilon_u\beta_u + D_u^2\sigma_\delta^2}$ with the boundary values of Table 1, for every quadrupole setting K that satisfies the match. With $a = 19.35$ mm this gives 17.455 mm/18.865 mm at the line entrance (m1) and 18.555 mm/19.064 mm at the exit (x/y), exactly the values found numerically for both settings in the four-station model. The two interior stations have 30 mm halfwidths and could limit only if a five-sigma size exceeded about 12.5 mm, which no matched setting approaches. In this four-station model the nominal

minimum clearance is thus analytically independent of K: the "saturation" of the minimum is a consequence of placing restrictions only where the optics are fixed, not a finding about the optics. This statement concerns the nominal, error-free, exactly matched line. Under errors the exit covariance $\Sigma_{\text{out}} = M(K,e)\Sigma_{\text{in}}(e)M(K,e)^{\mathsf{T}}$ also depends on K, through gradient errors, rolls, chromatic transport of off-momentum particles and the sampled entrance beta factors; together with the centroid response, these make the exit clearance K-dependent. The same nominal statement holds for any restriction downstream of q33, whereas restrictions upstream of q11 are independent of K even with errors. All error-study results below transport the full perturbed covariance and centroid of each error realization.

At a one-sided edge a_+, such as the septum blade, $c_u = a_+ - \bar{u} - 5\sigma_u$ with $\sigma_u = \sqrt{\Sigma_{uu}}$, and the local sensitivity to a quadrupole strength is

$$\frac{\partial c_u}{\partial K_j} = -\frac{\partial \bar{u}}{\partial K_j} - \frac{5}{2\sigma_u}\frac{\partial \Sigma_{uu}}{\partial K_j}. \tag{5}$$

At a symmetric aperture the first term carries the sign of $\bar{u}$; switching restrictions or crossing zero centroid makes the expression piecewise. In an uncoupled plane $\Sigma_{uu} = \epsilon_u\beta_u + D_u^2\sigma_\delta^2$; our coupled calculation retains the complete covariance. Reducing beta controls one size contribution, not necessarily dispersion or orbit sensitivity. We export these variances and centroids separately, adding variances rather than RMS sizes.

3.2 Jitter transfer between matched ends

For the error-free, uncoupled line with both ends exactly matched, the horizontal transfer matrix from m1 to m4 is

$$M_x = B_4 R(\mu_x) B_1^{-1}, \qquad B = \begin{pmatrix} \sqrt{\beta} & 0 \\ -\alpha/\sqrt{\beta} & 1/\sqrt{\beta} \end{pmatrix}, \tag{6}$$

with $R(\mu_x)$ a rotation by the horizontal phase advance and $B_{1,4}$ the boundary Courant–Snyder parameters of Table 1. Let J be the covariance of the non-dispersive part of the shot centroid at m1 (position 0.5 mm and angle 0.2 mrad rms, uncorrelated); the momentum jitter $\sigma_{\delta,j}$ is added separately through the dispersion. The centroid therefore reaches the exit with covariance $M_x J M_x^{\mathsf{T}}$ plus the momentum term $\sigma_{\delta,j}^2 \mathbf{d}_4\mathbf{d}_4^{\mathsf{T}}$, which is the same for every matched setting. In this nominal model the quadrupoles influence the exit jitter only through μ_x. Over a full period the exit position spread ranges from 0.26 mm to 0.95 mm and the angle spread from 0.12 mrad to 0.41 mrad, in opposite phase (Fig. 4, left): a matched line can move booster jitter between position, which the septum blade sees, and angle, which the injection kicker sees, but cannot remove it. On error realizations, the residual trajectory after steering, the optics errors and roll coupling add to this jitter; the test losses are computed with the full transport of each realization (Sections 4.2 and 4.3). Averaged exactly over the jitter, these static contributions raise the blade loss of the error-free line by only a few percent (Section 4.5); the larger tracked losses of Table 6 reflect the jitter tail sampled by the 600 test shots.

3.3 Loss-based observable and design

For a Gaussian projection with centroid $\bar{u}$ and rms size σ at an aperture with edges $a_\pm$, the lost fraction is

$$P = \Phi\left(\frac{a_- - \bar{u}}{\sigma}\right) + \Phi\left(\frac{\bar{u} - a_+}{\sigma}\right), \tag{7}$$

with Φ the standard normal distribution function. Equation (7) is exact in each plane for the rectangular and one-sided apertures, including the septum blade. For the round apertures of the pipe and quadrupoles we evaluate instead the exact probability that the Gaussian (x, y) distribution lies outside the circle, using a closed-form radial integral and periodic angular quadrature. Two projected tails would underestimate it: for a centered isotropic beam at a

circle of radius 5σ the exact loss is $e^{-12.5} = 3.73 \times 10^{-6}$, whereas $4\Phi(-5) = 1.15 \times 10^{-6}$. Per shot, the line loss is bracketed by the largest of these exact event probabilities and, as an upper bound, by their sum; we report the former as the prediction and use it as the design surrogate. The blade at the injection-septum entrance and at the line end gives the largest event in every evaluated case, and every tracked first loss occurs in the injection septum. When the centroid is itself Gaussian, as for linearly transported jitter, the shot-averaged lost fraction is Eq. (7) with the summed covariance of beam and jitter. This makes a smooth, loss-relevant objective. The loss-based setting minimizes the logarithm of the largest shot-averaged P on the error-free line, subject to the six exit equalities, horizontal peak beta $\leq 60\,\mathrm{m}$ and the strength bounds, using sequential least-squares programming (SLSQP) [17] with a 200-iteration budget and termination tolerance 10^{-10}. Three initial settings are evaluated in the bounded search. The objective uses the nominal entrance beta functions and the error-free line; the static residuals and the 5 % beta-function fluctuations of the test are not part of it, so the error-free prediction and the tracked test describe different populations. The setting is frozen before a fresh test on 200 steered error realizations, evaluated with tracked particles. Paired differences between settings are averaged per error realization; the pointwise 95 % intervals are percentile intervals of 10 000 bootstrap resamples of the 200 realizations. This realization-cluster bootstrap accounts for the dependence between the shots of one realization, but it does not certify that the heavy loss tail is converged; it describes the sampled population of realizations and shots.

3.4 Accuracy and spatial observation resolution

Independent hard-edge maps reproduce AT tracking at every element boundary within 1.6×10^{-8}, against a tolerance of 2×10^{-6} in the stated SI convention. On the checked error cases, centered-map, integration-step, probe-step and quadrature checks change clearances by at most 0.6858 µm, far below the reported effects, and the sampled peak beta functions change by less than 0.35 mm between 32 and 64 subdivisions per element. These are finite-case checks, not universal error bounds or loss-tail certificates; their details, including the audit of the earlier four-station campaign, are given in the supplement. In the low-beta setting the second quadrupole of the third triplet, `q32`, remains at its lower bound $-3\,\mathrm{m}^{-2}$.

4 Results

4.1 Active restrictions of the element-wise line

Figures 2 and 3 compare the baseline setting with the optimized low-beta setting. The baseline strengths are the quadrupole settings supplied with the BTS configuration before this study; no record states the exit target for which they were designed, and they do not satisfy the present one (Table 3). They are therefore a starting point, not a competing design. Their nominal five-sigma envelope extends across the injection-septum blade boundary, but this is expected: the blade distance is defined from the matched exit beam (Section 2.3), so any setting with a much larger exit beta function overlaps it. The optimized setting satisfies the exit target, reduces the large vertical envelope, and restores positive nominal clearance. Table 3 gives the corresponding exit lattice functions, peak beta values and minimum clearances. The figure comparison therefore includes the improvement from matching the baseline BTS line, rather than holding the exit optics fixed for both curves.

After matching, a separate comparison of the matched reference and optimized low-beta setting isolates the effect of interior beta reduction. In that matched comparison the blade clearances are 0.648 mm at the injection-septum entrance and 0.683 mm at the line end (m4), essentially equal for both settings. The nearest interior restriction is the vertical gap of b2, with 9.12 mm for the matched reference and 9.32 mm for the optimized low-beta setting. The bend-gap and extraction-septum aperture sweeps leave the blade as the limiting restriction for

Table 3: Nominal optimization of the baseline BTS setting. Peak beta functions are sampled with the same spatial resolution as Figure 2. The maximum normalized exit residual is evaluated using the declared optical scales; a matched setting requires it to be below 10^{-3}. A negative clearance means that the projected five-sigma envelope overlaps an aperture boundary, not that every particle is lost.

Observable	Baseline setting	Optimized low beta
Exit β_x [m]	44.9808	2.3365
Exit β_y [m]	242.6069	4.2562
Peak β_x [m]	52.2506	33.7945
Peak β_y [m]	242.6069	28.4371
Minimum clearance [mm]	-1.5170	0.6484
Max. normalized exit residual	4.77×10^{3}	2.88×10^{-10}

these matched settings. Thus optimization of the baseline BTS line improves nominal delivery, while further reductions of interior beta at a fixed exit target act mainly on clearance away from the blade. The error-study tables below retain this matched reference as a separate comparison setting; they do not report an error ensemble for the baseline setting.

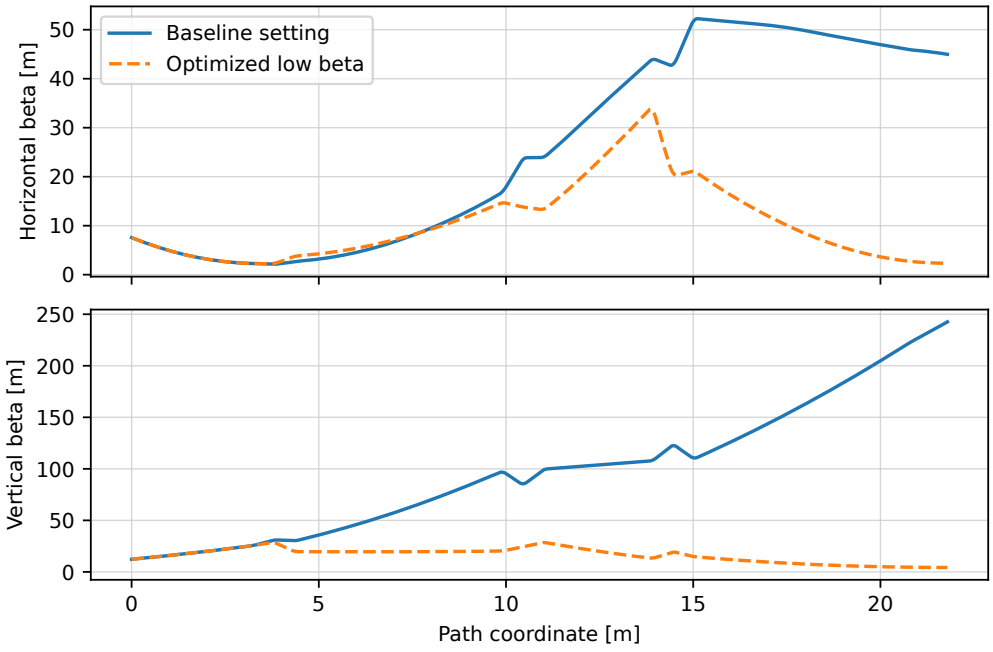


Figure 2: Baseline setting and optimized low-beta lattice functions, drawn with Accelerator Toolbox linear-optics plotting from the same entrance Courant–Snyder parameters and with the same observation resolution. The baseline setting uses the unmodified configuration strengths and does not satisfy the exit target; the optimized setting is matched. Both curves use the same magnet positions and bend geometry.

4.2 Trajectory correction

Table 4 compares the uncorrected and corrected matched lines on 200 error realizations with three shots of 2 000 particles each; both settings and both variants see the same magnet errors, shots and particles. Without correction the static errors displace the exit centroid by 1.2 mm rms,

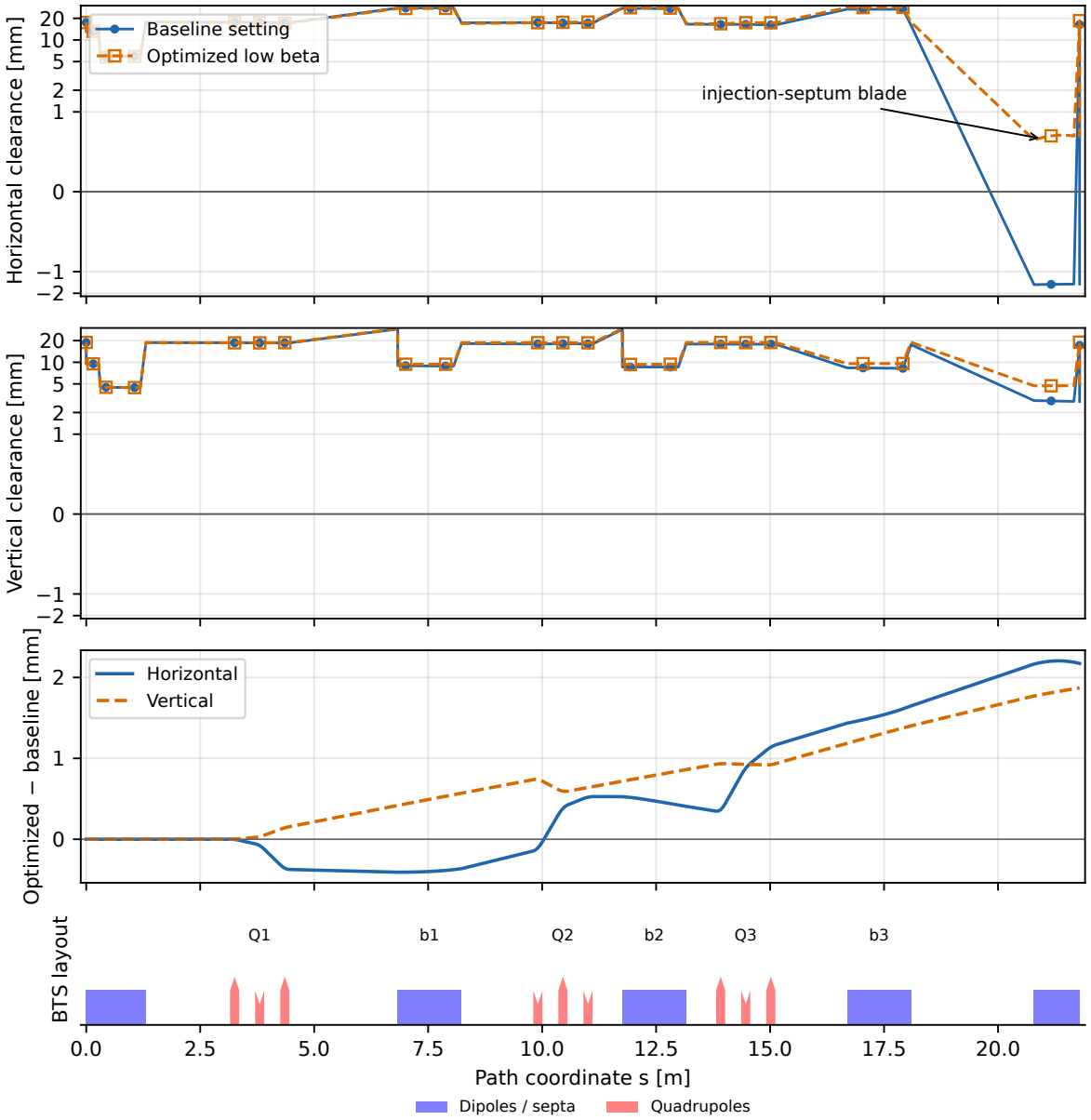


Figure 3: Nominal five-sigma clearance for the baseline setting (solid blue with filled circles) and optimized low beta (dashed orange with open squares). A symmetric logarithmic scale, linear near zero, displays the baseline setting's negative horizontal clearance at the injection septum. Zero marks the five-sigma envelope touching an aperture. The third panel shows optimized minus baseline clearance in each plane. The bottom panel gives the common BTS layout; Q1–Q3 identify quadrupole triplets. The baseline and optimized settings share the entrance conditions but have different exit lattice functions. Design beam, no errors.

and 10.08 % of the particles are lost; every recorded first loss is at the injection-septum blade. With the dipole trims the exit position spread falls to 684 µm, the angle spread stays at 367 µrad (364 µrad uncorrected), and the loss falls to 1.68 % (1.47 % for the low-beta setting); the rms trim is 204 µrad. Two trims close to the exit, b3 and the injection septum, steer both position and angle at the septum. The uncorrected vertical exit centroid is 1116 µm and 100 µrad rms for the matched reference (Table 5); it stays inside the 5 mm vertical restrictions but is the static vertical injection error passed to the injection calculation.

Table 4: Steering of the matched reference (Matched) and optimized low beta on 200 error realizations (3 shots, 2 000 particles each). $\bar{c}_{\min}$ is the mean minimum five-sigma clearance, Lost the tracked particle-loss fraction, Predicted the mean Gaussian loss probability of the most loss-prone aperture, Blocks the number of error realizations with any observed loss, and x_{rms}, x'_{rms} the rms exit centroid over shots.

Setting	Steering	$\bar{c}_{\min}$ [mm]	Lost [%]	Predicted [%]	Blocks	x_{rms} [mm]	x'_{rms} [mrad]
Matched	none	0.584	10.08	10.04	74	1.20	0.36
Low beta	none	0.556	10.00	10.01	79	1.20	0.38
Matched	dipole trims	0.524	1.68	1.68	47	0.68	0.37
Low beta	dipole trims	0.523	1.47	1.47	44	0.68	0.38

Two observations arise from these results. First, after correction the exit centroid spread, about 0.68 mm and 0.37 mrad, is mainly that of the booster jitter (0.64 mm and 0.33 mrad for the error-free matched reference, Table 6); averaged exactly over the jitter, the static residual raises the loss by only a few percent (Section 4.5). Second, the mean minimum clearance does not rank the variants by loss: the uncorrected line has the largest mean clearance (0.584 mm against 0.524 mm) and six times the loss. All 600 shots enter both means, and the moments are uncensored by apertures, so this is not a survivor effect. Without correction the static errors leave a mean exit offset of −68.5 µm, away from the blade (8.8 µm after correction), which raises the mean clearance, while the spread of the exit centroid grows from 0.68 mm to 1.20 mm. At a one-sided edge a with fixed beam size σ and centroid spread τ, the mean clearance $a - E[\bar{u}] - 5\sigma$ does not depend on τ, whereas the mean loss, approximately $\Phi\big(-(a - E[\bar{u}])/\sqrt{\sigma^2 + \tau^2}\big)$, grows with it; the same mean clearance can therefore hide very different tail losses. The mean Gaussian loss prediction, by contrast, agrees with the tracked loss fraction of each row within 0.4 % relative. Because the tracked beam is itself Gaussian, this agreement checks the moment transport and loss accounting; it does not test non-Gaussian booster tails, which would raise the blade loss most where the blade is many beam sizes away. After correction the low-beta setting loses 0.21 percentage points less than the matched reference on these realizations (paired interval [−0.39, −0.06] percentage points), whereas its mean clearance change, −0.001 mm ([−0.009, 0.007] mm), is unresolved. On the fresh test of Section 4.3 its loss change is unresolved. The small difference follows its phase advance (Section 4.3), not its lower peak beta.

Table 5 repeats the steering on the same realizations with fewer readings and with b1–b3 trimmed as one series-powered knob, the kick of each bend proportional to its bending angle. The full layout reproduces the tracked losses above exactly. Reading only the exits of the three triplets, the septum entrance and the line end leaves the losses unchanged to within 0.01 percentage point. Series-powered bends leave one knob fewer near the exit: the exit angle spread grows from 0.37 mrad to 0.44 mrad for the matched reference, and the loss of the loss-based setting rises from 0.23 % to 0.31 %, still well below that of the matched reference.

4.3 Phase advance of the matched settings and the loss-based setting

Bounded least-squares matching from 360 random starts finds 349 matched quadrupole settings within the strength bounds, with horizontal phase advances between 200.2° and 232.0°. To test

Table 5: Steering layouts on the 200 realizations of Table 4 (tracked, 3 shots of 2 000 particles). Readings: quadrupole exits (all nine or the three triplet exits), septum entrance and line end. Knobs: six independent dipole trims, or four with b1–b3 in series. Exit centroids are rms over shots; the vertical plane is not corrected.

Readings / knobs	Setting	Lost [%]	Pred. [%]	x_{rms} [mm]	x'_{rms} [mrad]	y_{rms} [mm]	y'_{rms} [mrad]
11 / 6	Matched	1.68	1.68	0.68	0.37	1.116	0.100
	Loss-based (R3)	0.23	0.23	0.54	0.42	0.765	0.094
5 / 6	Matched	1.67	1.68	0.68	0.37	1.116	0.100
	Loss-based (R3)	0.23	0.23	0.54	0.42	0.765	0.094
11 / 4	Matched	1.65	1.65	0.69	0.44	1.115	0.100
	Loss-based (R3)	0.31	0.31	0.55	0.51	0.765	0.095
5 / 4	Matched	1.58	1.59	0.69	0.44	1.115	0.100
	Loss-based (R3)	0.31	0.31	0.55	0.51	0.765	0.095

whether this range reflects the search or the matching problem, we scanned μ_x in 2° steps as a seventh matching condition, and searched for the smallest and largest μ_x subject to the six exit equalities, the strength bounds and peak beta functions of at most 60 m in both planes, from five starts per edge (Fig. 5). Matched settings are found on the grid from 202° to 232°. The lower edge, 200.2°, is reached by three converged searches with no strength bound and neither peak-beta limit active, so it is set by the matching conditions themselves within the bounds, as far as these local searches show; the upper edge, 232.0°, is set by the 60 m limit in both planes. These are ranges found by sampled and local searches, not a proof of global reachability. The matched reference (211.8°) and low-beta (210.8°) settings lie near the middle. As Eq. (6) implies, the predicted blade loss of the error-free line depends on the setting through μ_x alone: it falls monotonically from 4 % at the upper edge to 0.27 % at the lower edge, while the peak beta functions vary independently between about 23 m and 60 m. Reducing the loss therefore means choosing the smallest matchable μ_x.

The loss-based search of the earlier study ended at that edge, $\mu_x = 200.2°$, although all three of its SLSQP starts reached the 200-iteration limit and only one was feasible. The converged edge solution has the same phase advance to within 8×10^{-7}° and the same peak beta to within 0.02 m; we therefore regard the loss-based setting as the best-found feasible setting at the lower edge of the matched range, not as a certified global optimum. Its predicted shot-averaged loss is 0.27 % against 1.28 % for the matched reference, because its exit position jitter is 506 µm instead of 643 µm (Table 6).

Table 6: Matched reference (Matched), optimized low-beta and loss-based settings and their fresh test (200 steered error realizations, 3 shots, 2 000 particles each). μ_x: horizontal phase advance; $\hat{\beta}$: larger peak beta; $\sigma_{x,j}$, $\sigma_{x',j}$: exit jitter from Eq. (6); $\bar{P}$: shot-averaged loss on the error-free line; Lost and Pred.: tracked and predicted loss on the test realizations; Blocks: error realizations with any loss.

Setting	μ_x [deg]	$\hat{\beta}$ [m]	$\sigma_{x,j}$ [mm]	$\sigma_{x',j}$ [mrad]	$\bar{P}$ [%]	Lost [%]	Pred. [%]	Blocks
Matched	211.8	51.3	0.643	0.327	1.28	2.30	2.31	57
Low beta	210.8	33.8	0.631	0.331	1.16	2.27	2.28	55
Loss-based	200.2	48.1	0.506	0.368	0.27	0.69	0.69	29

On the fresh error realizations the tracked loss is 2.30 %, 2.27 % and 0.69 % for the matched reference, low-beta and loss-based settings, each within 1 % relative of its mean prediction. The loss-based setting changes the loss by -1.61 percentage points against the matched reference (pointwise 95 % interval clustered by error realization, [-2.33, -0.96] percentage points), a factor of 3.3; low beta changes it by -0.03 percentage points ([-0.23, +0.17] percentage points), unresolved. Its mean minimum clearance changes by only −0.017 mm ([-0.035, +0.001] mm), also unresolved:

the mean-clearance metric cannot resolve this loss reduction. The price is an exit angle jitter larger by 13 % (368 µrad against 327 µrad), which the injection kicker must accept. The loss is concentrated in shots whose entrance jitter is two to three standard deviations: the five worst of the 200 test realizations of the matched reference contribute 38 % of its loss, the worst losing 43 % per shot on average. The intervals, clustered by error realization, account for this heavy tail, and the population mean differs between independently seeded populations (Section 4.5).

On a further 200 independently seeded realizations (Table 7), the tracked loss is 0.97 % for the matched reference, 0.86 % for low beta and 0.13 % for the converged edge solution, a change of -0.84 [-1.39, -0.40] percentage points (a factor of 7.5). The earlier loss-based setting loses 0.09 % on the same realizations although it is optically identical to the edge solution: their jitter-integrated losses, 0.28 % and 0.29 %, agree, and the tracked difference comes from the monitor reading errors, which are drawn separately for each setting (Section 2.4), and from the sampled jitter tail. On this population low beta also loses resolvably less than the matched reference (-0.11 [-0.24, -0.01] percentage points), again through its slightly smaller μ_x. The loss-based gain thus ranges from a factor of 3 to more than 10 between populations, while its sign is the same in every population tested.

Table 7: Second fresh test (200 steered error realizations with new seeds, 3 shots, 2 000 particles each). $\bar{P}$: predicted loss of the error-free line; Integr.: jitter-integrated loss on the steered realizations; Lost: tracked loss; last column: paired difference from the matched reference with its pointwise 95 % realization-cluster interval. The loss-based (R3) and phase-edge settings are optically identical; Disp.-free is the best dispersion-free scan setting.

Setting	μ_x [deg]	$\hat{\beta}$ [m]	$\sigma_{x',j}$ [mrad]	$\bar{P}$ [%]	Integr. [%]	Lost [%]	Lost − Matched [pp]
Matched	211.8	51.3	0.327	1.28	1.32	0.97	—
Low beta	210.8	33.8	0.331	1.16	1.20	0.86	-0.11 [-0.24, -0.01]
Loss-based (R3)	200.2	48.1	0.368	0.27	0.28	0.09	-0.89 [-1.47, -0.42]
Scan grid best	202.0	37.4	0.362	0.37	0.38	0.25	-0.73 [-1.20, -0.35]
Phase edge	200.2	48.1	0.368	0.27	0.29	0.13	-0.84 [-1.39, -0.40]
Disp.-free	188.0	35.6	0.393	4.7×10^{-3}	6.2×10^{-3}	8.3×10^{-5}	-0.97 [-1.64, -0.45]

4.4 Exit target and dispersion

At m4 the configured target gives the injected beam an rms divergence of 76 µrad: 57 µrad from the betatron emittance and 50 µrad from $D'_x\sigma_\delta$. The 0.1 % shot energy jitter adds a centroid angle of 47 µrad and a position of 81 µm rms. Both terms are fixed by the target, the same for every matched setting, and absent for a dispersion-free exit.

The configured exit dispersion is that of the injected beam in the configuration of the NKM injection study; we have found no derivation of it from the ring optics, whose dispersion at the injection kicker is nearly zero. Matching the same Courant–Snyder parameters with $D_x = D'_x = 0$ at m4 is feasible within the strength bounds: all 115 solutions from 120 bounded random starts keep both peak betas below 60 m (smallest larger peak 26.3 m). The constrained searches place the dispersion-free range between 187.2° and 214.5° (Fig. 5); its upper edge is set by q22 at its strength bound. At the lower edge the searches that reach 187.2° stopped at their iteration limit with no constraint active, and the one converged search ends at 187.6° on the vertical 60 m limit. The range includes phases where booster jitter falls mainly on the exit angle. The best scanned setting ($\mu_x = 188°$, larger peak beta 35.6 m) has a predicted error-free blade loss of 4.7×10^{-3} % and an exit angle jitter of 393 µrad; the beam divergence falls to 57 µrad. Tracked on the second test population, it loses 1 of 1.2 million particles, consistent with its jitter-integrated prediction of 6.2×10^{-3} %; at such small losses non-Gaussian booster tails, not modeled here, would dominate.

Among the alternatives tested here, the exit target therefore has the largest leverage on injection. Removing the exit dispersion lowers the injected beam's divergence by a quarter and

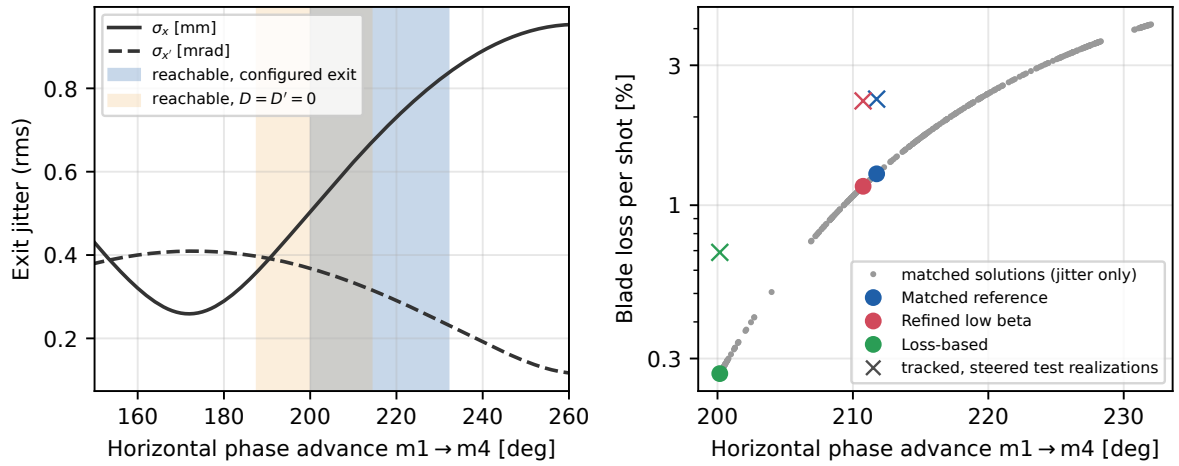


Figure 4: Left: rms exit position (solid, mm) and angle (dashed, mrad) jitter from Eq. (6) versus horizontal phase advance, with the phase-advance ranges found for the configured exit (blue) and a dispersion-free exit (orange; its momentum term differs slightly). Right: predicted shot-averaged blade loss of every matched solution (grey) and of the three settings (circles), with the tracked loss on steered test realizations (crosses), which also include static residuals. The matched reference in this comparison is the intermediate matched setting; the baseline setting appears in Figures 2 and 3.

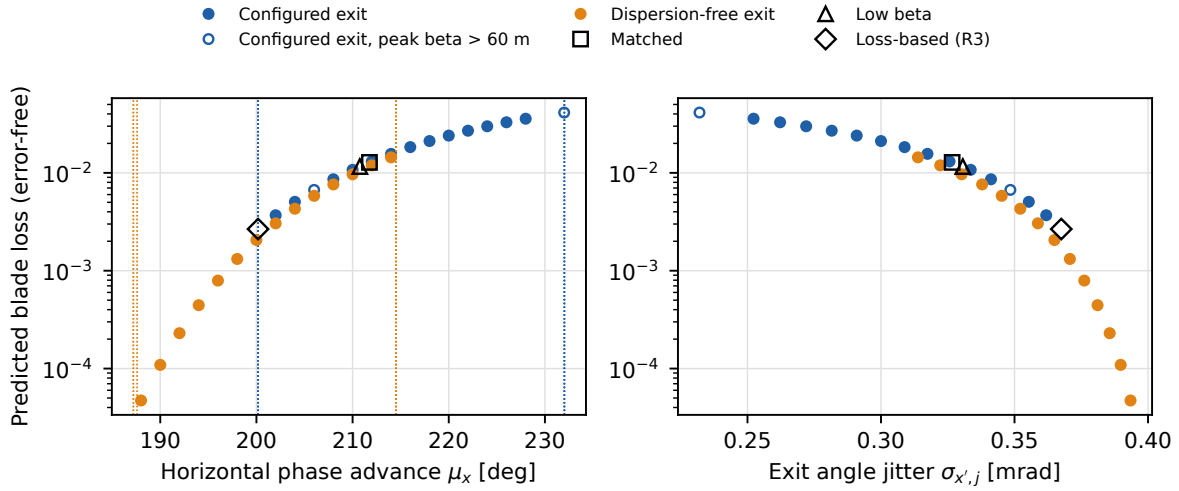


Figure 5: Phase-advance scan. Left: predicted blade loss of the error-free line for the lowest-peak-beta matched setting at each scanned μ_x, for the configured exit (blue) and a dispersion-free exit (orange); open symbols exceed the 60 m peak-beta limit. Dotted lines mark the edges found by constrained searches. Right: the same settings against the exit angle jitter. Black symbols mark the matched reference, low-beta and loss-based settings.

lets the line put booster jitter where the septum blade does not see it, but the angle jitter, 394 µrad against 327 µrad for the matched reference, then arrives at the injection kicker. In the NKM injection study [5], the kicker's 80 % angular acceptance is 0.44 mrad, and once the static vertical injection error left by transfer-line magnet errors is removed, booster position and angle jitter are the dominant kicker-specific capture losses. This line has no vertical correction; correcting that vertical error will be addressed in a future study. Whether the trade favors the blade or the kicker must be decided with the ring, not within the line.

4.5 Sensitivity to the error levels

The 0.5 mrad rms quadrupole roll is large for a transfer line, and 0.1 % rms shot energy jitter is at the high end of booster performance. Table 8 repeats the steered evaluation with rolls of 0.1 mrad, 0.2 mrad and 0.5 mrad and energy jitter of 0.05 % and 0.1 % on another 200 independently seeded error realizations (analytic loss and clearance, no particle tracking). Changing the roll level changes the predicted loss by less than 0.001 percentage points, i.e. not at all at the precision reported in Table 8: quadrupole rolls couple a small part of the horizontal motion into the vertical plane, whose restrictions keep several millimeters of clearance, and leave the horizontal blade loss unchanged. Halving the energy jitter lowers the matched-reference loss from 1.34 % to 1.32 %, because its 81 µm exit contribution is small compared with the betatron jitter. The ranking of the settings is the same at every roll and energy-jitter level. On this population the loss-based setting is 5.6 times below the matched reference, against 3.3 on the tracked test population: with three shots per realization, the size of the gain depends on the sampled jitter tail.

For each steered realization we therefore also evaluate the jitter-integrated loss: Eq. (7) averaged exactly over the Gaussian entrance jitter on that realization's steered line, which resolves losses far below the reach of sampled shots. Table 9 scales the position and angle jitter together on a further 200 independently seeded realizations. At the assumed level the jitter-integrated losses are 1.31 % and 0.28 % for the matched reference and loss-based settings, close to the error-free 1.28 % and 0.27 %: the steered static residual adds only 2 % and 5 %, and the loss-based gain is 4.7. Halving the jitter lowers the losses by more than two orders of magnitude, with a gain of 31. At 0.1–0.2 of the assumed level, i.e. 13–44 % of the beam size at m1, every setting loses a fraction of at most 3×10^{-11}, and the loss-based setting is no better than the others; for it, the static residual then contributes more than the jitter. The loss reduction reported above is therefore a property of large booster jitter. The partition of the jitter between exit position and angle (Section 3.2) holds at every level, because it scales linearly with the jitter.

Table 8: Sensitivity of the steered matched line (Matched denotes the matched reference) to the roll and energy-jitter levels: mean predicted loss and mean minimum five-sigma clearance on 200 error realizations with three shots each (analytic, not tracked). Each row holds for all three roll levels, whose results are identical at the printed precision.

Roll	$\sigma_{\delta,j}$	Predicted loss [%]			$\bar{c}_{\min}$ [mm]	
[mrad]	[%]	Matched	Low beta	Loss-based	Matched	Loss-based
0.1, 0.2, 0.5	0.05	1.32	1.20	0.23	0.519	0.504
0.1, 0.2, 0.5	0.10	1.34	1.24	0.24	0.517	0.502

The centroid-to-blade distance $d_b = 1.478$ mm is taken from the NKM injection geometry, and the blade loss depends on it roughly exponentially. Figure 6 repeats the jitter-integrated evaluation on the same 200 steered realizations for d_b from 1 mm to 3 mm. At the assumed jitter, a loss below 0.1 % requires $d_b \geq 2.05$ mm for the matched reference, 2.01 mm for low beta and 1.64 mm for the loss-based setting; at $d_b = 1$ mm the losses are 6.6 % and 3.0 %. With a tenfold smaller jitter the loss stays below 2×10^{-6} even at 1 mm. A larger d_b places the injected beam farther from the stored beam and so demands more of the ring's dynamic acceptance at injection,

Table 9: Dependence on the booster betatron jitter. Position and angle jitter at m1 are scaled together from the assumed 0.5 mm and 0.2 mrad (scale 1) and are given in units of the rms beam size and divergence at m1; the momentum jitter stays at 0.1 %. Losses are the jitter-integrated predictions averaged over 200 steered error realizations, and the last column is the error-free line (analytic, not tracked).

Jitter	At m1 [beam sizes]		Jitter-integrated loss, steered [%]			
scale	position	angle	Matched	Low beta	Loss-based	Error-free matched
0	0.00	0.00	2.7×10^{-13}	3.8×10^{-13}	2.5×10^{-10}	3.4×10^{-14}
0.05	0.07	0.11	4.4×10^{-13}	5.7×10^{-13}	2.8×10^{-10}	4.9×10^{-14}
0.1	0.13	0.22	2.8×10^{-12}	2.7×10^{-12}	3.8×10^{-10}	3.0×10^{-13}
0.2	0.26	0.44	2.6×10^{-9}	1.9×10^{-9}	1.3×10^{-9}	7.9×10^{-10}
0.5	0.66	1.10	3.2×10^{-3}	2.3×10^{-3}	1.0×10^{-4}	2.6×10^{-3}
1	1.32	2.21	1.3	1.2	2.8×10^{-1}	1.3

which must be evaluated with the injection calculation [5]. A thinner blade, such as the 1 mm septum of the TDR design [2], would either add 1 mm to d_b at the same injection offset or allow the same d_b with the injected beam 1 mm closer to the stored beam.

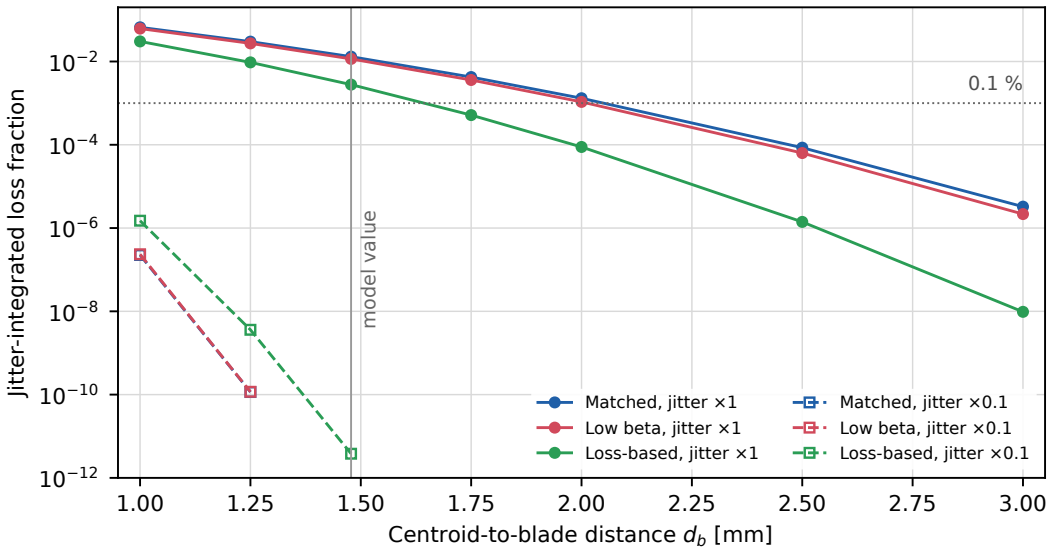


Figure 6: Jitter-integrated blade loss against the centroid-to-blade distance d_b for the three matched settings at the assumed booster jitter (solid, filled) and a tenfold smaller jitter (dashed, open), averaged over 200 steered error realizations (analytic). Values below 10^{-12} are omitted. The vertical line marks the model value 1.478 mm and the dotted line a 0.1 % loss.

5 Discussion and conclusions

Matching the initial setting to the prescribed exit is a prerequisite rather than a result of this study: the blade distance is defined from the matched exit beam, so a setting whose exit beta function is far from the target necessarily overlaps the blade. The questions addressed here start from the matched settings and concern what still differs between them once the endpoint optics are fixed.

In a BTS line matched at both ends, the nominal beam size at the ends is fixed and so is the nominal clearance of every restriction outside the quadrupole section. With the blade geometry

of the NKM injection study, the injection septum is that restriction, and it limits the line by an order of magnitude more than any interior aperture. Lowering the interior peak beta, the usual design heuristic, therefore changes only clearances of several millimeters and, on fresh steered error realizations, leaves the loss unchanged.

The quadrupoles act on the loss mainly through the trajectory. Static trajectory errors are reduced by field trims of the existing dipoles, without adding any element to the line; the two trims nearest the exit steer both position and angle at the septum. A horizontal residual, the optics errors and the uncorrected vertical trajectory remain. The dominant remaining contribution is booster shot jitter, and in the nominal matched line its arrival at the exit depends only on the horizontal phase advance. The line chooses whether the jitter appears as position at the blade or as angle at the injection kicker, but it cannot remove it. For the configured exit, the matched settings found span about 32° of phase advance; the error-free blade loss depends on μ_x alone and is smallest at the lower edge of that range, which no strength bound or peak-beta limit sets. The best-found setting there reduces the tracked loss by a factor of 3 to more than 10 on independently seeded test populations, at 13 % more angle jitter. A dispersion-free exit extends the range to smaller μ_x and lowers the loss much further, again at the cost of angle jitter. The choice between blade loss and injection-angle jitter belongs to the injection design as a whole: for the NKM, whose 80 % angular acceptance is 0.44 mrad, exit-angle jitter of 0.33 mrad–0.39 mrad is of the same order, so the trade must be evaluated with ring capture, and the booster's angle stability bounds what either choice can achieve.

Two methodological points carry over. The mean minimum five-sigma clearance does not rank settings by loss: it rewards static offsets away from a one-sided restriction and is blind to the jitter tail. A Gaussian loss probability from the propagated centroid and covariance agrees with the mean tracked losses within 1 % relative, is smooth, and can be optimized directly. The phase-advance view also gives a quick test for any matched transfer line: compute μ_x, read off where the booster jitter lands, and compare with the restriction and the kicker acceptance.

The booster jitter assumed here is 1.3 and 2.2 beam sizes at m1, and the blade loss and the benefit of the loss-based setting depend on it steeply: at 0.1–0.2 of that level every matched setting loses less than 10^{-10}, so the design choice that remains is where the jitter appears at the exit, not how much is lost at the blade. With sampled shots the loss is heavy-tailed and its population mean varies between independently seeded populations. The dispersion-free results are analytic. Losses are counted at the declared aperture check points (element entrances and the line end), which bracket the septum channel but do not resolve continuous interception inside it. This paper neither certifies real transmission nor derives construction tolerances, ring capture, or kicker transparency.

Statements and Declarations

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References

[1] R. Hettel. DLSR design and plans: an international overview. *J. Synchrotron Rad.*, 21:843–855, 2014. https://doi.org/10.1107/S1600577514011515.

[2] Pohang Accelerator Laboratory. Korea-4GSR technical design report. Technical Report 4GSR-I-DE-TDR-001, Pohang Accelerator Laboratory, October 2025. https://kls.kbsi.re.kr/eng/sub0305/view/id/63794.

[3] J. Kim, G. Jang, J. Kim, J. Lee, and J. Seok. The Korea-4GSR storage ring lattice design. In *Proc. 16th Int. Particle Accelerator Conf. (IPAC'25)*, pages 789–792, Taipei, Taiwan, 2025. https://doi.org/10.18429/JACoW-IPAC2025-MOPS089.

[4] Y. Lee, E.-S. Kim, S. H. Park, S. Shin, and J. Kim. Study on beam injection and ramping efficiency for Korea-4GSR booster synchrotron. In *Proc. 15th Int. Particle Accelerator Conf. (IPAC'24)*, pages 1025–1028, Nashville, TN, USA, 2024. paper TUPC12; https://doi.org/10.18429/JACoW-IPAC2024-TUPC12.

[5] Chong Shik Park. Finite-length and vertical effects in nonlinear-kicker injection into a Korea Light Source design lattice. Manuscript in preparation, 2026.

[6] Y. Y. Guo, Z. Duan, Y. Jiao, Y. M. Peng, and G. Xu. High energy transport line design for the HEPS project. In *Proc. IPAC'17*, pages 1466–1468, Copenhagen, Denmark, 2017. https://proceedings.jacow.org/ipac2017/papers/tupab063.pdf.

[7] Yuanyuan Guo, Yuanyuan Wei, Yuemei Peng, and Gang Xu. The transfer line design for the HEPS project. *Radiation Detection Technology and Methods*, 4(4):440–447, 2020. https://doi.org/10.1007/s41605-020-00209-6.

[8] I. P. S. Martin, R. T. Fielder, J. Kallestrup, T. Olsson, B. Singh, J. K. Jones, and B. Muratori. Beam dynamics studies for the Diamond-II injector. In *Proc. IPAC'22*, pages 2708–2711, Bangkok, Thailand, 2022. https://accelconf.web.cern.ch/ipac2022/papers/thpopt049.pdf.

[9] ESRF. EBS Storage Ring Technical Report. Technical report, 2018. Accessed 13 October 2026; pp. 285–286, Fig. 9.19; https://www.esrf.fr/files/live/sites/www/files/com%20photos/About%20Us/EBS-TECHREPORT-reduced.pdf.

[10] J. Kallestrup and V. Schlott. Beam control and characterization of the new SLS 2.0 booster-to-ring transfer line. In *Proc. IPAC'25*, 2025. https://proceedings.jacow.org/ipac2025/pdf/TUPM056.pdf.

[11] X. R. Resende, M. B. Alves, F. H. de Sá, L. Liu, A. C. S. Oliveira, and J. V. Quentino. Sirius injection optimization. In *Proc. IPAC'22*, pages 2672–2674, 2022. https://proceedings.jacow.org/ipac2022/papers/thpopt038.pdf.

[12] S. C. Leemann. Pulsed sextupole injection for Sweden's new light source MAX IV. *Phys. Rev. ST Accel. Beams*, 15:050705, 2012. https://doi.org/10.1103/PhysRevSTAB.15.050705.

[13] Hao-Wen Luo, Fan-Hsin Tseng, Mau-Sen Chiu, Nuan-Ya Huang, and Ping J. Chou. Design of a new booster to storage ring transfer line for nonlinear kicker injection scheme in Taiwan Photon Source. IPAC'26, Deauville, France, contribution TUP2685, 2026. Conference abstract; https://indico.jacow.org/event/95/contributions/14036/.

[14] A. Terebilo. Accelerator modeling with MATLAB accelerator toolbox. In *Proc. 2001 Particle Accelerator Conf. (PAC'01)*, pages 3203–3205, Chicago, IL, USA, 2001. paper RPAH314.

[15] W. Rogers, N. Carmignani, L. Farvacque, and B. Nash. pyAT: A python build of accelerator toolbox. In *Proc. 8th Int. Particle Accelerator Conf. (IPAC'17)*, pages 3855–3857, Copenhagen, Denmark, 2017. https://doi.org/10.18429/JACoW-IPAC2017-THPAB060.

[16] Andrey N. Tikhonov and Vasiliy Y. Arsenin. *Solutions of Ill-Posed Problems*. V. H. Winston & Sons, Washington, DC, 1977.

[17] Dieter Kraft. A software package for sequential quadratic programming. Technical Report DFVLR-FB 88-28, DFVLR, Institut für Dynamik der Flugsysteme, Oberpfaffenhofen, Germany, 1988. `https://degenerateconic.com/uploads/2018/03/DFVLR_FB_88_28.pdf`.